Brief Overview of the course & Human and Environment

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Course Layout

Topic 1: Ecology and Environment
Topic 2: Environmental Pollutants
Topic 3: Water Pollution
Topic 4: Air Pollution
Topic 5: Waste Management
Topic 6: Noise Pollution
Topic 7: Environmental Impact Assessment

All classes start from 10 minutes of Centering and relaxation to re-orientate ourselves in the class.

Important resource materials

Resource 1: Environmental Engineering by Howard S. Peavy, Donald R. Rowe, George Tchobanoglous; McGraw Hill Education.

Assignments and interactive sessions

Assignment 1: Video preparation of environmental awareness
Assignment 2: Periodical surprise quizzes

Reference: Odisha, a whirlwind of terror | India 2050, https://www.youtube.com/watch?v=SbQFEFRibCA, Accessed on: 31 Dec, 2024

DO YOU THINK ANYWHERE AT INDIVIDUAL LEVEL THESE HAVOCS CAN BE REDUCED

- Waste pollution
- Global warming
- Drinking Water pollution
- Faecal Sludge pollution
- Groundwater pollution
- Air pollution

Sustainable Development

Meeting the needs of the present without compromising the ability of the future generations to meet their own needs.

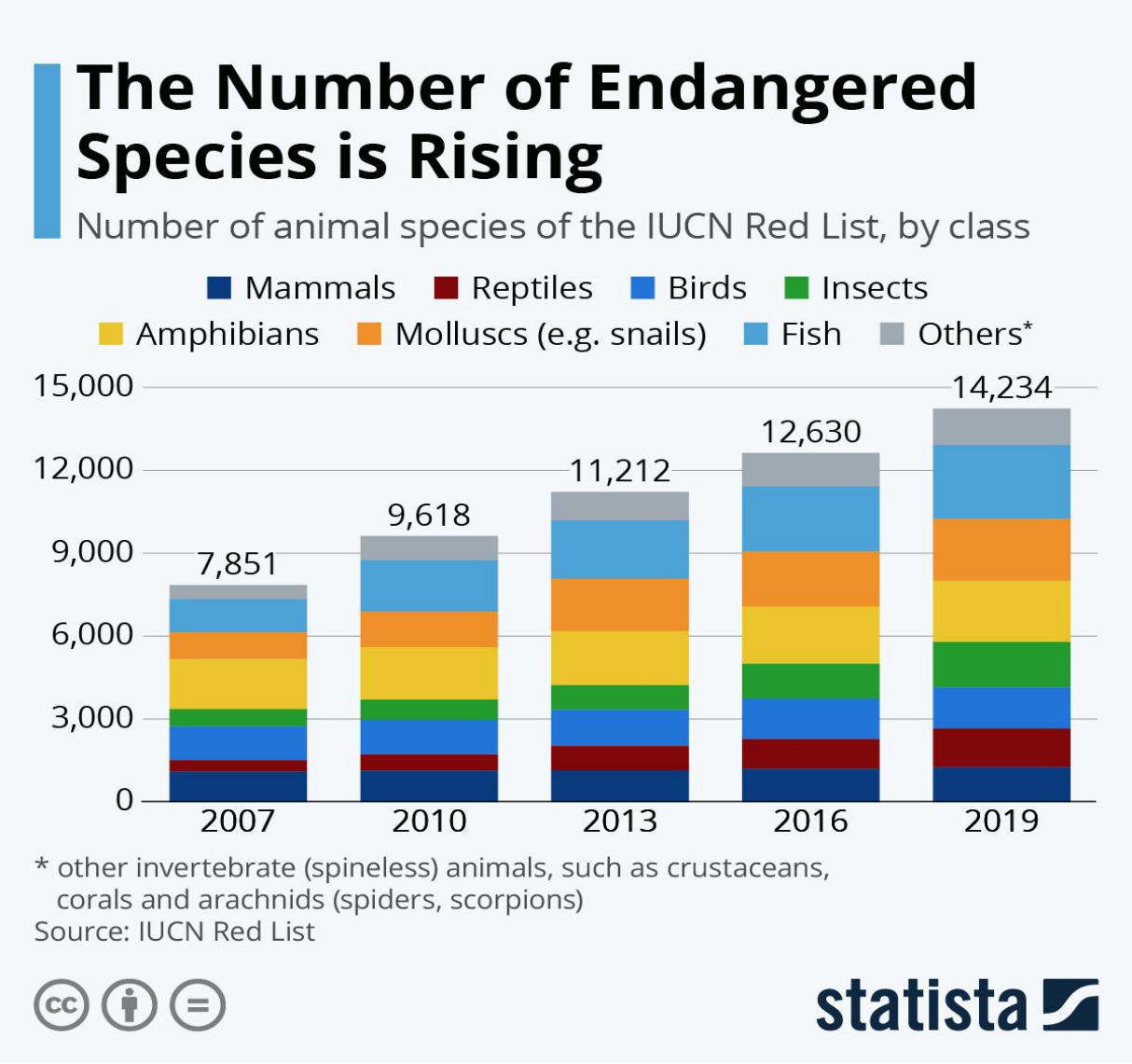
Source: 1. United Nations World Commission on Environment and Development, 1987, 2. Boredpanda, 25 comics of sustainable development, https://www.boredpanda.com/environmental-comics-inspiring-to-take-action-rewriting-extinction/ Accessed on: 3 Jan, 2025

Source: United Nations World Commission on Environment and Development, 1987

FACT CHECK

- Extinction is a natural phenomenon, it occurs at a natural "background" rate of about one to five species per year. Scientists estimate we're now losing species at 1,000 to 10,000 times the "background" rate, with dozens going extinct every day.

- In 2020, the figure has reached 18,829 incorporating 1299 Mammals, 1406 Reptiles, 1486 birds, 1819 insects, 2276 Amphibians, 2275 Molluscs, 2849 Fishes and 5419 others.

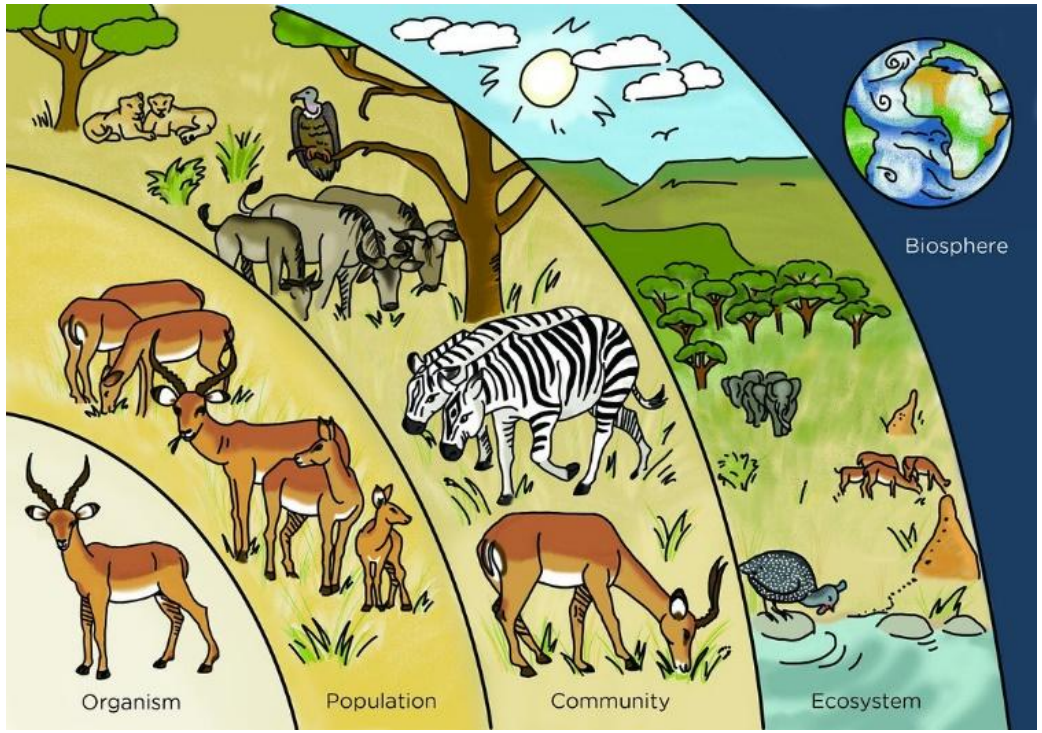


Source: https://www.statista.com/chart/17122/number-of-threatened-species-red-list/
file:///C:/Users/Lenovo/Downloads/2020-2_RL_Stats_Table1a.pdf
https://www.dosomething.org/us/facts/11-facts-about-endangered-species#fnref6
https://www.iucnredlist.org/resources/summary-statistics

What is Ecology?

- Ecology is derived from the Greek word **eco that means "house" and logy means "study", this means "the study of one's house"**.

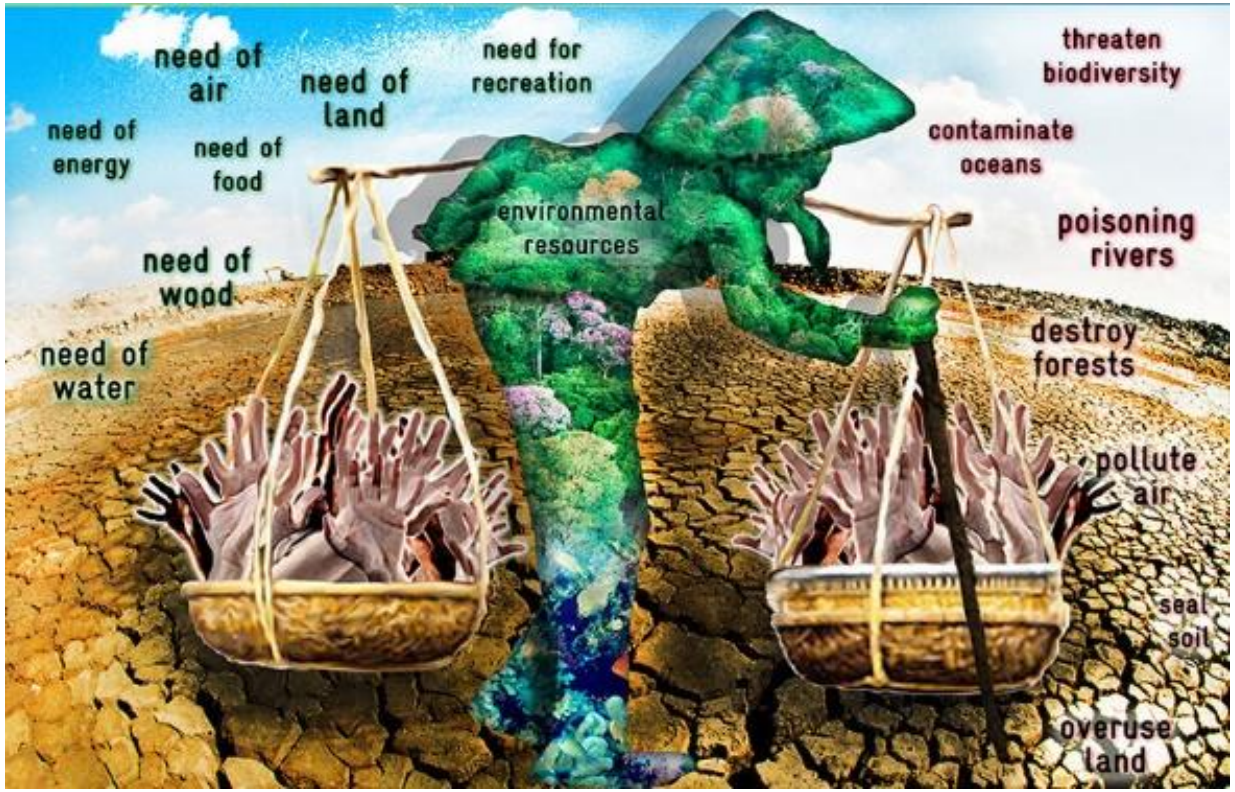
- It is the study of systems that include interactions among and between the organisms and also their interaction with non living Environment.



Source: 1. EIA Lecture notes of Dr. T.V. Ramachandra (IISC Bangalore) 2. Bing Search engine, CCBY

What is Environment?

- Environment is the surroundings that affect an organism during its lifetime.



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Time for Adventure

Heartfulness, Does human has anything to learn from Nature? (https://youtu.be/wsX7BMH-d-E)

Thank you

For any questions, please feel free to reach out to me.

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